

Futures Pricing

Models And Comparisons

FINM 37000 - Futures And Related Derivatives

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Argument by No-arbitrage

Deducing present value (in)equalities

Consider two static (no position change) portfolios with time t value P_t and R_t .

No-arbitrage principle: If $P_T \geq R_T$ at fixed future time T , then $P_t \geq R_t$ at all earlier times t .

Why? How would you profit if $P_t < R_t$ at some earlier time t ?

Corollary: If $P_T = R_T$ at fixed future time T , then $P_t = R_t$ at all earlier t .

Forward Price By No-Arbitrage

Financing a spot position

Consider forward contract to buy a non-dividend paying stock with price S_t

Forward contract struck at K has time T value: $S_T - K$

What assets are available to match time T payoff (replicating portfolio)?

- Risk free asset paying a continuous rate r for maturity T : e^{rT}
- The stock: 1 unit will have value S_T
- The forward: 1 forward will have value $S_T - K$

Forward Price By No-Arbitrage

Financing a spot position

- Time T:
 - P_T : Long 1 asset: S_T , short K of the bond
 - R_T : Long 1 forward
 - Both have value $S_T - K$
- Time 0 value for the same portfolios
 - P_0 : Stock S_0 , bond $e^{-rT}K$
 - R_0 : Forward value: 0
 - By no-arbitrage: $S_0 - e^{-rT}K = 0$

$$K = S_0 e^{rT} = F_0$$

- Forward price reflects asset growth at the risk-free rate

You have X forwards expiring at time T entered at time 0 with forward price F_0 . At time t , you want to exit your position when the forward price is F_t . How would you value your time t P&L? Hint: when do you realize this P&L?

0

$$XF_t$$

0%

$$XF_0e^{rt}$$

0%

$$X(F_t - F_0)$$

0%

$$X(F_t - F_0)e^{-r(T-t)}$$

0%

$$X(F_t - F_0)e^{-rt}$$

0%

$$X(F_T - F_t)e^{-r(T-t)}$$

0%

Futures Price By No-arbitrage

Offsetting with a future

- Sell a future at time 0 for price F_0 , value 0, to offset the financed purchase of the asset
- No-arbitrage argument is tough with margin and static replication
 - Model-dependent expected no-arbitrage values
 - Usually use forward as a good enough approximation

Futures Pricing

Daily Cashflow Accounting For One Long Future

- R_t is the 1-day interest return on the margin account
- t is measured in days in the next few slides

t	Futures Price	Futures Value	Margin
0	F_0	0	M_0
$0 < t < 1$	F_t	$F_t - F_0$	M_0
$t \rightarrow 1^-$ (pre-settle)	F_1	$F_1 - F_0$	$M_0 R_0$
$t \rightarrow 1^+$ (settle)		0	$M_0 R_0 + F_1 - F_0$
2^-			
2^+			
...			
$t \rightarrow T^-$			
$t \rightarrow T^+$		0	

You have X futures expiring at time T entered at time 0 with futures price F_0 . At time t during the same day, you want to exit your position when the futures price is F_t . What is your time t realized P&L?



$$XF_t$$

0%

$$XF_0e^{rt}$$

0%

$$X(F_t - F_0)$$

0%

$$X(F_t - F_0)e^{-r(T-t)}$$

0%

$$X(F_t - F_0)e^{-rt}$$

0%

$$X(F_T - F_t)e^{-r(T-t)}$$

0%

Futures Pricing

T=1 day Futures Price By No-Arbitrage

Assume bond and margin pay the same constant interest rate.

Portfolio: Buy asset, offset by short future, finance with bonds. How many bonds?

t	Future	Spot	Margin	Bond
0	0	S_0	M_0	X
$t \rightarrow T^-$	$F_T - F_0$	S_T	M_0	X
$t \rightarrow T^+$	0	S_T	$M_0 e^{rT} + F_T - F_0$	$X e^{rT}$

Futures Pricing

T=1 day Futures Price By No-Arbitrage

$t = 0$	0	S_0	M_0	X
$t \rightarrow T^+$	0	S_T	$M_0e^{rT} + F_T - F_0$	Xe^{rT}

Long asset, short future, short bonds

1. Express T value, $F_T \rightarrow S_T$:

$$S_T + M_0e^{rT} - S_T + F_0 - Xe^{rT} = M_0e^{rT} + F_0 - Xe^{rT}$$

2. Pick X to set replicating relationship:

$$M_0e^{rT} + F_0 - Xe^{rT} = 0 \implies X = F_0e^{-rT} + M_0$$

3. By no-arbitrage, at $t = 0$,

$$S_0 + M_0 - F_0e^{-rT} - M_0 = 0 \implies F_0 = S_0e^{rT}$$

Futures Pricing

T=2 days Futures Price By No-Arbitrage

Assume bond and margin pay the same constant interest rate.

Portfolio: Buy asset, offset by short futures, finance with bonds.

How many bonds? How many futures?

t	Futures Quantity	Futures Value	Spot	Margin	Bond
0	Y_0	0	S_0	M_0	X
$t \rightarrow 1^+$	Y_1	0	S_1	$M_0e^r + Y_0(F_1 - F_0)$	Xe^r
$t \rightarrow T^-$	Y_1	$F_T - F_0$	S_T	$M_0e^r + Y_0(F_1 - F_0)$	Xe^r
$t \rightarrow T^+$	0	0	S_T	$M_0e^{rT} + Y_1(F_T - F_1) + e^rY_0(F_1 - F_0)$	Xe^{rT}

Futures Pricing

T=2 days Futures Price By No-Arbitrage

0	Y_0	0	S_0	M_0	X
$t \rightarrow T^+$	0	0	S_T	$M_0 e^{rT} + Y_1(F_T - F_1) + e^r Y_0(F_1 - F_0)$	$X e^{rT}$

1. Express T value, $F_T \rightarrow S_T$:

$$S_T + M_0 e^{rT} - Y_1(S_T - F_1) - e^r Y_0(F_1 - F_0) - X e^{rT}$$

2. Pick Y_0, Y_1 to make the F_1, S_T terms cancel. $Y_0 =$ $Y_1 =$

3. Pick X to set replicating relationship:

$$M_0 e^{rT} + F_0 - X e^{rT} = 0 \implies X = F_0 e^{-rT} + M_0$$

4. By no-arbitrage, at $t = 0$,

$$S_0 + M_0 - F_0 e^{-rT} - M_0 = 0 \implies F_0 = S_0 e^{rT}$$

Futures Pricing

General T

- Assume bond and margin pay the same constant interest rate. Then futures price and forward price are both
$$F_0 = S_0 e^{rT}$$
- What happens when the interest rate is not constant?
 - Think first: what if F_t and r_t move together? What happens to your margin?
 - Use stochastic models to estimate the size of the effect.

Cost-of-carry Model

Moving beyond non-dividend paying stocks

- Borrow to buy asset for S_0
- Holding until time T leads to having asset worth S_T and owing
$$V_{0,T} = S_0 + I_{0,T} + C_{0,T} - B_{0,T}$$
- $I_{0,T}$: interest paid on borrowing
- $C_{0,T}$: additional costs specific to the asset esp. storage
- $B_{0,T}$: benefits from the asset (e.g., dividends, interest payments)

Cost-of-carry Model

Offsetting with a forward

- Sell a forward at time 0 for price F_0 (not value) to offset the financed purchase of the asset
- At time $t = 0$, forward has value 0, asset $+S_0$, borrowing $-S_0$
- At time $t = T$,
 - Forward value $S_T - F_0$, so short position is $F_0 - S_T$
 - Total position value $(F_0 - S_T) + S_T - V_{0,T}$
 - No arbitrage implies $F_0 - V_{0,T} \leq 0$, that is

$$F_0 \leq S_0 + I_{0,T} + C_{0,T} - B_{0,T}$$

Cost-of-carry Model

Bounding Forward Price From Below

- When can we reverse the cost-of-carry trade?
 - Need to be able to short the asset
 - Typically would be responsible for paying benefits like dividends
 - Typically do not need to offset the buyer's costs (e.g., they store it)

$$F_0 \geq S_0 + I_{0,T} - B_{0,T}$$

- If costs are negligible, $F_0 = V_{0,T}$ by considering both bounds.
- For many contracts, costs are not negligible!

Cost-of-carry Model

Discrete/Accrued Convenience Yield

Difference between forward prices and the cost-of-carry is termed the (accrued) convenience yield $Y_{0,T} \geq 0$:

$$F_0 = S_0 + I_{0,T} + C_{0,T} - B_{0,T} - Y_{0,T}$$

Positive convenience yield reflects the desirability of holding the asset beyond the value already reflected in S_0 and $B_{0,T}$.

Cost-of-carry Model

Continuous Convenience Yield

y : the continuous convenience yield

r : the continuous interest rate

u : the continuous rate of non-interest costs

q : the rate of benefits

c : the cost of carry

$$F_0 = S_0 e^{(r+u-q-y)T} = S_0 e^{(c-y)T}$$

WARNING: Sometimes $c - y$ is called the cost-of-carry, and other times just c , i.e., the costs that are not part of the convenience yield. These symbols are not standardized. Check your sources.

It is also common to assume these rates are constant, but that is an additional assumption. There is really an implicit dependence on the period $[0, T]$ for all of these rates.

Basis Trade

Spot vs. Futures or Forwards

- Basis $F_t - S_t$ (or $S_t - F_t$ in some market conventions)
- Key to trading opportunities (the basis trade):
 - Understanding costs and benefits now and in the future
 - Data on storage costs, dividends, etc.
 - Understanding factors affecting convenience yield
 - Expectations about future supply and demand
 - Delivery optionality
- Access to execution across markets (e.g., physical markets, FX markets, bond markets, equity markets)

Basis Risk

Spot vs. Futures or Forwards

- Hedging S_t exposure with futures (basis risk)
- Types of basis risk
 - Product risk: futures underlying may not match asset
 - Examples
 - Crude oil futures as a hedge for refined products
 - ES futures as hedge for equity market risk
 - Timing risk
 - futures expirations may not match timing of risk exposure
 - Optionality in the timing if delivery is taken
 - Location risk: physical delivery may require additional transport